	Scotch Student ID #			
Circle the relevant digits	0	0	0	0
	1	1	1	1
	2	2	2	2
	3	3	3	3
	4	4	4	4
	5	5	5	5
	6	6	6	6
	7	7	7	7
	8	8	8	8
	9	9	9	9

Teacher's Name

Scotch College

CHEMISTRY

SAC 5 – Chemistry of food & Chromatography

Date: Thursday 8th September, 2022

Reading Time	0 minutes
Writing Time	45 minutes

Task Sections	Marks	Your Marks
Total Marks	45	

General Instructions

- Write your student number in the space provided above on this page.
- Please complete all questions in the spaces provided.

Allowed Materials

- A VCAA Chemistry Data Book has been provided.
- Students are permitted to bring into the SAC: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator.

At the end of the task…

- Please ensure your student number has been filled in at the top of the page and that you have written your teacher's name.
- You may keep the data book.

Electronic Devices

Students are **not** allowed to have a mobile phone, smart watch and/or any other unauthorised electronic device in the SAC, unless it is TURNED OFF and is placed on the front teacher desk.

Q1.

In response to a pain stimulus, the brain produces small polypeptide molecules. These molecules block the transmission of pain through the central nervous system. The amino acid sequence in one such molecule is:

Met – Ala – Asp — Lys – Ile

The number of amino, carboxy and amide functional groups within this sequence of the polypeptide is:

	Amino	**Carboxy**	**Amide**
A	1	1	5
B	2	2	5
C	1	1	4
D	2	2	4

Q2.

When a protein is heated to 70°C its secondary and tertiary structure changes. Which of the following bond types is least likely to be disrupted during this heating process?

A Dipole-dipole bonds

B Peptide linkages

C Hydrogen bonds

D Dispersion forces

Q3.

A bomb calorimeter, with a calibration factor of 2.25 kJ $°C^{-1}$ was used to burn a sample of 2.05 g sample of food which contained 45.0% m/m of fat and 55.0% m/m protein. Calculate the temperature in which the calorimeter would rise, in °C.

A 23.2

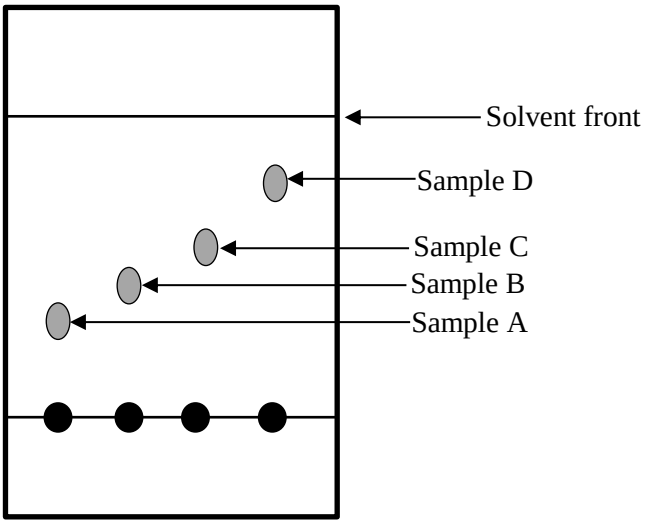
B 23.7

C 15.2

D 20.2

Q4. .

The thin layer chromatography plate shown below has a polar stationary phase. It was developed using hexane as the solvent.



Which sample has the least polar molecules?

A Sample A

B Sample B

C Sample C

D Sample D

Q5.

The number of carbon-to-carbon double bonds in a molecule can be identified by reacting the molecule, (such as a triglyceride or fatty acid) with hydrogen gas. If 17.50 g of a fatty acid becomes fully saturated with exactly of 3.10 L of H_2 gas at SLC, determine the identity of the fatty acid from the list below.

A Linoleic (280 g/mol)

B Arachidic (312 g/mol)

C Arachidonic (304 g/mol)

D Oleic (282 g/mol)

Q6.

Which of the following is most likely to be a fatty acid obtained by the hydrolysis of a triglyceride from food?

A $C_3H_8O_3$

B $C_{12}H_{22}O_{11}$

C $C_6H_8O_6$

D $C_{16}H_{32}O_2$

Q7. (10 marks)

The table below shows the nutritional information of dried chia seeds. Chia seeds are considered to be low GI and rich in many vitamins and minerals.

	Per 100 g of dried chia seeds
Proteins	16.5 g
Carbohydrates	42.1 g (including 80 % insoluble dietary fibre)
Saturated fat	3.33 g
Monounsaturated fats	2.31 g
Polyunsaturated fats	23.7 g

a) Calculate the energy available to the human body, in kJ, of 100.0 g serving of chia seeds. (4 marks)

b) Name the carbohydrate which makes up the insoluble fibre in chia seeds. (1 mark)

c) Using your knowledge of glycaemic index, (GI), explain why chia seeds are known to be low GI, compared to foods such as white bread which is considered to be high GI.

In your response you should

- provide a definition of GI
- discuss both amylose and amylopectin. (3 marks)

d) Chia seeds are often used as a source of protein in our diet. An athlete, with a mass 72.0 kg requires 85.0 g of protein each day, to maintain their muscle mass.

Assuming 60.0% of the available protein in chia seeds is metabolised into a usable form, calculate the mass of chia seeds that would need to be consumed per day to meet the protein requirements for this athlete.

(2 marks)

Q8. (8 marks)

The table below contains a number of fatty acids which can be found in the data book.

Fatty acid	**Formula**	**Heat of combustion (kJ/mol)**
Lauric	$C_{11}H_{23}COOH$	7415
Palmitoleic	$C_{15}H_{29}COOH$	10031
Stearic	$C_{17}H_{35}COOH$	12500
Linoleic	$C_{17}H_{31}COOH$	11840
Linolenic	$C_{17}H_{29}COOH$	11506

a) What type of omega fatty acid is linolenic acid? (1 mark)

_____

b) Using the information in the table above, write a thermochemical equation for the complete combustion of solid lauric fatty acid at SLC.

(2 marks)

c) Which of the two fatty acids, stearic or linoleic, would you expect to have the highest melting point? Explain your answer with reference to structure and bonding.

(3 marks)

_____

_____

_____

_____

_____

d) A chemist was trying to determine the heat of combustion of another fatty acid, arachidonic fatty acid, $C_{19}H_{31}COOH$. They combusted exactly 10.25 g in an enclosed calorimeter containing 2.00 kg of water. The water increased in temperature from 21.5°C to 27.8°C.

Calculate the energy content of the fatty acid in kJ/g. (2 marks)

_____

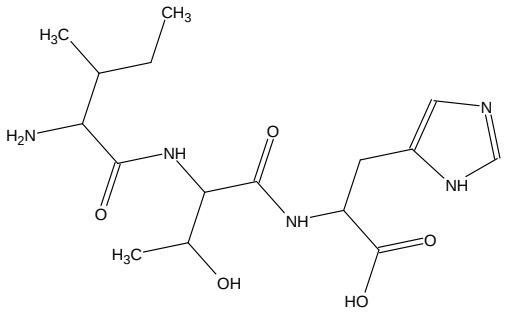
_____

_____

_____

Q9. (8 marks)

Below is a structure of a tripeptide.



a) Write the three-letter symbol of the **three** amino acids which comprise this structure above. (3 marks)

_____

b) Draw the full structural formula for the amino acid methionine, at neutral pH. (2 marks)

c) Using the tripeptide structure shown above, circle all groups which are responsible for creating the secondary structure in a protein.

(1 mark)

d) Describe the secondary structure of a protein with reference to bonding and its overall shape. (2 marks)

_____

_____

_____

_____

e) Identify two **different** side chain interactions that could occur between two of the tripeptide molecules shown above. Write these below. (2 marks)

_____

_____

Q10. (8 marks)

Students were analysing vitamin C and D content in a protein shake. A sample of the protein drink was passed through a **reverse phase HPLC,** with a polar mobile phase and a non-polar stationary phase and their retention times recorded.

The data is shown below.

Molecule	**Retention time (min)**
Molecule A	2.50
Molecule B	8.65

a) Based on the retention of the molecules above, which molecule A or B, would you expect to be vitamin D?

(1 mark)

_____

b) Justify your answer in Q10 a) with reference to structure and bonding of vitamin D and vitamin C

(2 marks)

_____

_____

_____

_____

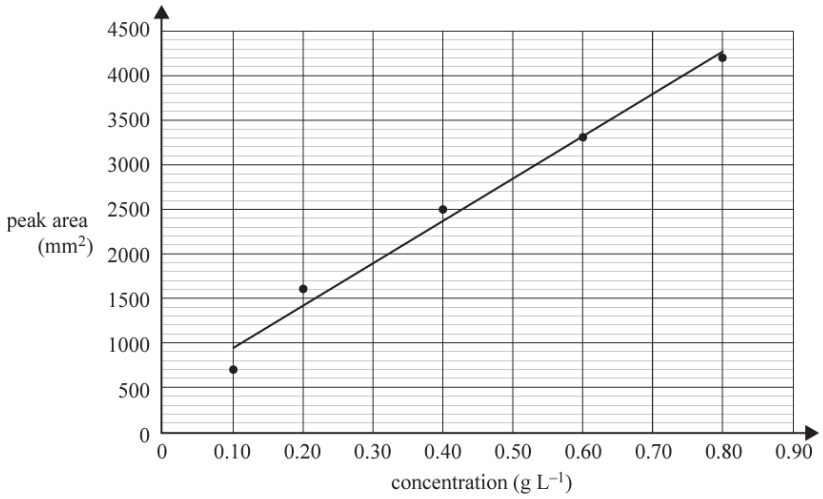
_____

c) Identify one change that could be made to the conditions of the HPLC machine, so that there would be an increased separation between molecules A and B. (1 mark)

_____

_____

In another experiment, students wanted to determine the sugar content of a protein shake. Various standards of sucrose were made to produce using a calibration curve, and their peak area recorded, as shown below.



d) Subsequently a 375.0 mL carton of a protein shake was analysed for its sucrose content, using the following procedure.

A 25.00 mL sample of the protein shake was taken and placed into a 250.0 mL volumetric flask, where water was then added up to the mark.

This solution was then injected into the HPLC column with the peak area producing a peak area of 3300 mm^2.

i. Using this information and the calibration curve above, calculate the amount (in grams per litre) of the sucrose in the sample that was injected into the HPLC.

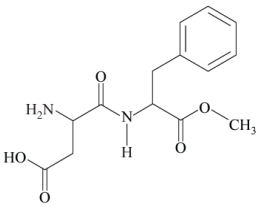
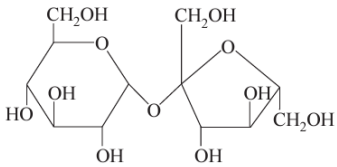
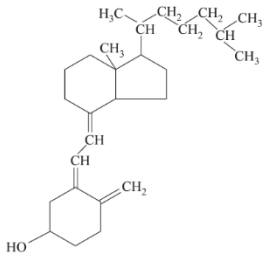
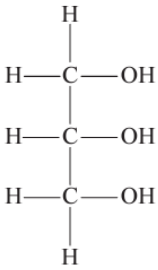
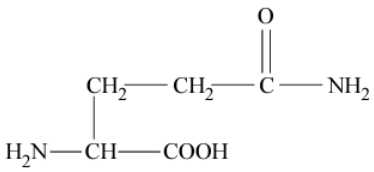
(1 mark)

ii. Calculate the percentage mass/volume (%m/v) of sucrose in the 25.00 mL sample of the soft drink (2 marks)

iii. The sample of sucrose was sourced from a 375.0 mL protein shake. Assuming no other sweeteners were present, calculate the mass of sucrose, in grams, in the original carton. (1 mark)

Q11. (3 marks)

Examine the following molecules.

Molecule A	Molecule B	Molecule C
		
Molecule D	**Molecule E**	**Molecule F**
	$C_{15}H_{29}COOH$	

Circle all the letters(s) that correspond to structures matching each of the following descriptions (A – F).

Statement	**Molecules**					
Contains a glycosidic linkage	A	B	C	D	E	F
Have both acid and base properties	A	B	C	D	E	F
Are considered largely hydrophobic	A	B	C	D	E	F

End of SAC – 45 marks available